12. We should note that $\overline{\mathbb{Q}(x)}$ is an algebraic closure of $\overline{\mathbb{Q}}(x)$. We know that π is transcendental over $\mathbb{Q}$. Therefore, $\sqrt{\pi}$ must be transcendental over $\mathbb{Q}$, for if it were algebraic, then $\pi = (\sqrt{\pi})^2$ would be algebraic over $\mathbb{Q}$, because algebraic numbers form a closed set under field operations. Therefore the map $\tau : \mathbb{Q}(\sqrt{\pi}) \to \mathbb{Q}(x)$ where $\tau(a) = a$ for $a \in \mathbb{Q}$ and $\tau(\sqrt{\pi}) = x$ is an isomorphism. Theorem 49.3 shows that τ can be extended to an isomorphism σ mapping $\overline{\mathbb{Q}(\sqrt{\pi})}$ onto a subfield of $\overline{\mathbb{Q}(x)}$. Then σ^{-1} is an isomorphism mapping $\sigma[\overline{\mathbb{Q}(\sqrt{\pi})}]$ onto a subfield of $\overline{\mathbb{Q}(\sqrt{\pi})}$ which can be extended to an isomorphism of $\overline{\mathbb{Q}(x)}$ onto a subfield of $\overline{\mathbb{Q}(\sqrt{\pi})}$. But because σ^{-1} is already onto $\overline{\mathbb{Q}(\sqrt{\pi})}$, we see that σ must actually be onto $\overline{\mathbb{Q}(x)}$, so σ provides the required isomorphism of $\overline{\mathbb{Q}(\sqrt{\pi})}$ with $\overline{\mathbb{Q}(x)}$.

13. Let E be a finite extension of F. Then by Theorem 31.11, $E = F(\alpha_1, \alpha_2, \cdots, \alpha_n)$ where each α_i is algebraic over F. Now suppose that $L = F(\alpha_1, \alpha_2, \cdots, \alpha_{k+1})$ and $K = F(\alpha_1, \alpha_2, \cdots, \alpha_k)$. Every isomorphism of L onto a subfield of $\overline{F}$ and leaving F fixed can be viewed as an extension of an isomorphism of K onto a subfield of $\overline{F}$. The extension of such an isomorphism τ of K to an isomorphism σ of L onto a subfield of $\overline{F}$ is completely determined by $\sigma(\alpha_{k+1})$. Let $p(x)$ be the irreducible polynomial for α_{k+1} over K, and let $q(x)$ be the polynomial in $\tau[K][x]$ obtained by applying τ to each of the coefficients of $p(x)$. Because $p(\alpha_{k+1}) = 0$, we must have $q(\sigma(\alpha_{k+1})) = 0$, so the number of choices for $\sigma(\alpha_{k+1})$ is at most $\deg(q(x)) = \deg(p(x)) = [L : K]$. Thus $\{L : K\} \leq [L : K]$, that is

$$\{F(\alpha_1, \cdots, \alpha_{k+1}) : F(\alpha_1, \cdots, \alpha_k)\} \leq [F(\alpha_1, \cdots, \alpha_{k+1}) : F(\alpha_1, \cdots, \alpha_k)]. \tag{1}$$

We have such an inequality (1) for each $k = 1, 2, \cdots, n-1$. Using the multiplicative properties of the index and of the degree (Corollaries 49.10 and 31.6), we obtain upon multiplication of these $n-1$ inequalities the desired result, $\{E : F\} \leq [E : F]$.

50. Splitting Fields

1. The splitting field is $\mathbb{Q}(\sqrt{3})$ and the degree over $\mathbb{Q}$ is 2.

2. Now $x^4 - 1 = (x-1)(x+1)(x^2+1)$. The splitting field is $\mathbb{Q}(i)$ and the degree over $\mathbb{Q}$ is 2.

3. The splitting field is $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ and the degree over $\mathbb{Q}$ is 4.

4. The splitting field has degree 6 over $\mathbb{Q}$. Replace $\sqrt[3]{2}$ by $\sqrt[3]{3}$ in Example 50.9.

5. Now $x^3 - 1 = (x-1)(x^2+x+1)$. The splitting field has degree 2 over $\mathbb{Q}$.

6. The splitting field has degree $2 \cdot 6 = 12$ over $\mathbb{Q}$. See Example 50.9 for the splitting field of $x^3 - 2$.

7. We have $|G(\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}| = 1$, because $\sqrt[3]{2} \in \mathbb{R}$ and the other conjugates of $\sqrt[3]{2}$ do not lie in $\mathbb{R}$ (see Example 50.9). They yield isomorphisms into $\mathbb{C}$ rather than automorphisms of $\mathbb{Q}(\sqrt[3]{2})$.

8. We have $|G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q}| = 6$, because $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$ is the splitting field of $x^3 - 2$ and is of degree 6, as shown in Example 50.9.

9. $|G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q}(\sqrt[3]{2})| = 2$, because $\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})$ is the splitting field of $x^2 + 3$ over $\mathbb{Q}(\sqrt[3]{2})$.

10. Theorem 33.3 shows that the only field of order 8 in $\overline{\mathbb{Z}_2}$ is the splitting field of $x^8 - x$ over $\mathbb{Z}_2$. Because a field of order 8 can be obtained by adjoining to $\mathbb{Z}_2$ a root of any cubic polynomial that is irreducible in $\mathbb{Z}_2[x]$, it must be that all roots of every irreducible cubic lie in this unique subfield of order 8 in $\overline{\mathbb{Z}_2}$.

11. The definition is incorrect. Insert "irreducible" before "polynomial".

Let $F \leq E \leq \overline{F}$ where $\overline{F}$ is an algebraic closure of a field F. The field E is a **splitting field over** F if and only if E contains all the zeros in $\overline{F}$ of every irreducible polynomial in $F[x]$ that has a zero in E.

12. The definition is incorrect. Replace "lower degree" by "degree one".

A polynomial $f(x)$ in $F[x]$ **splits in an extension field** E of F if and only if it factors in $E[x]$ into a product of polynomials of degree one.

13. We have $1 \leq [E : F] \leq n!$. The example $E = F = \mathbb{Q}$ and $f(x) = x^2 - 1$ shows that the lower bound 1 cannot be improved unless we are told that $f(x)$ is irreducible over F. Example 50.9 shows that the upper bound $n!$ cannot be improved.

14. T F T T T F F T T

15. Let $F = \mathbb{Q}$ and $E = \mathbb{Q}(\sqrt{2})$. Then $f(x) = x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3)$ has a zero in E, but does not split in E.

16. a. This multiplicative relation is not necessarily true. Example 50.9 and Exercise 7 show that $6 = |G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q})| \neq |G(\mathbb{Q}(\sqrt[3]{2}, i\sqrt{3})/\mathbb{Q}(\sqrt[3]{2}))| \cdot |G(\mathbb{Q}\sqrt[3]{2})/\mathbb{Q})| = 2 \cdot 1 = 2$.

b. Yes, because each field is a splitting field of the one immediately under it. If E is a splitting field over F then $|G(E/F)| = \{E : F\}$, and the index is multiplicative by Corollary 49.10.

17. Let E be the splitting field of a set S of polynomials in $F[x]$. If $E = F$, then E is the splitting field of x over F. If $E \neq F$, then find a polynomial $f_1(x)$ in S that does not split in F, and form its splitting field, which is a subfield E_1 of E where $[E_1 : F] > 1$. If $E = E_1$, then E is the splitting field of $f_1(x)$ over F. If $E \neq E_1$, find a polynomial $f_2(x)$ in S that does not split in E_1, and form its splitting field $E_2 \leq E$ where $[E_2 : E_1] > 1$. If $E = E_2$, then E is the splitting field of $f_1(x)f_2(x)$ over F. If $E \neq E_2$, then continue the construction in the obvious way. Because by hypothesis E is a *finite* extension of F, this process must eventually terminate with some $E_r = E$, which is then the splitting field of the product $g(x) = f_1(x)f_2(x) \cdots f_r(x)$ over F.

18. Find $\alpha \in E$ that is not in F. Now α is algebraic over F, and must be of degree 2 because $[E : F] = 2$ and $[F(\alpha) : F] = \deg(\alpha, F)$. Thus $\text{irr}(\alpha, F) = x^2 + bx + c$ for some $b, c \in F$. Because $\alpha \in E$, this polynomial factors in $E[x]$ into a product $(x - \alpha)(x - \beta)$, so the other root β of $\text{irr}(\alpha, F)$ lies in E also. Thus E is the splitting field of $\text{irr}(\alpha, F)$.

19. Let E be a splitting field over F. Let α be in E but not in F. By Corollary 50.6, the polynomial $\text{irr}(\alpha, F)$ splits in E since it has a zero α in E. Thus E contains all conjugates of α over F.

Conversely, suppose that E contains all conjugates of $\alpha \in E$ over F, where $F \leq E \leq \overline{F}$. Because an automorphism σ of $\overline{F}$ leaving F fixed carries every element of $\overline{F}$ into one of its conjugates over F, we see that $\sigma(\alpha) \in E$. Thus σ induces a one-to-one map of E into E. Because the same is true of σ^{-1}, we see that σ maps E onto E, and thus induces an automorphism of E leaving F fixed. Theorem 50.3 shows that under these conditions, E is a splitting field of F.

20. Because $\mathbb{Q}(\sqrt[3]{2})$ lies in $\mathbb{R}$ and the other two conjugates of $\sqrt[3]{2}$ do not lie in $\mathbb{R}$, we see that no map of $\sqrt[3]{2}$ into any conjugate other than $\sqrt[3]{2}$ itself can give rise to an automorphism of $\mathbb{Q}(\sqrt[3]{2})$; the other two maps give rise to isomorphisms of $\mathbb{Q}(\sqrt[3]{2})$ onto a subfield of $\overline{\mathbb{Q}}$. Because any automorphism of $\mathbb{Q}(\sqrt[3]{2})$ must leave the prime field $\mathbb{Q}$ fixed, we see that the identity is the only automorphism of $\mathbb{Q}(\sqrt[3]{2})$. [For an alternate argument, see Exercise 39 of Section 48.]